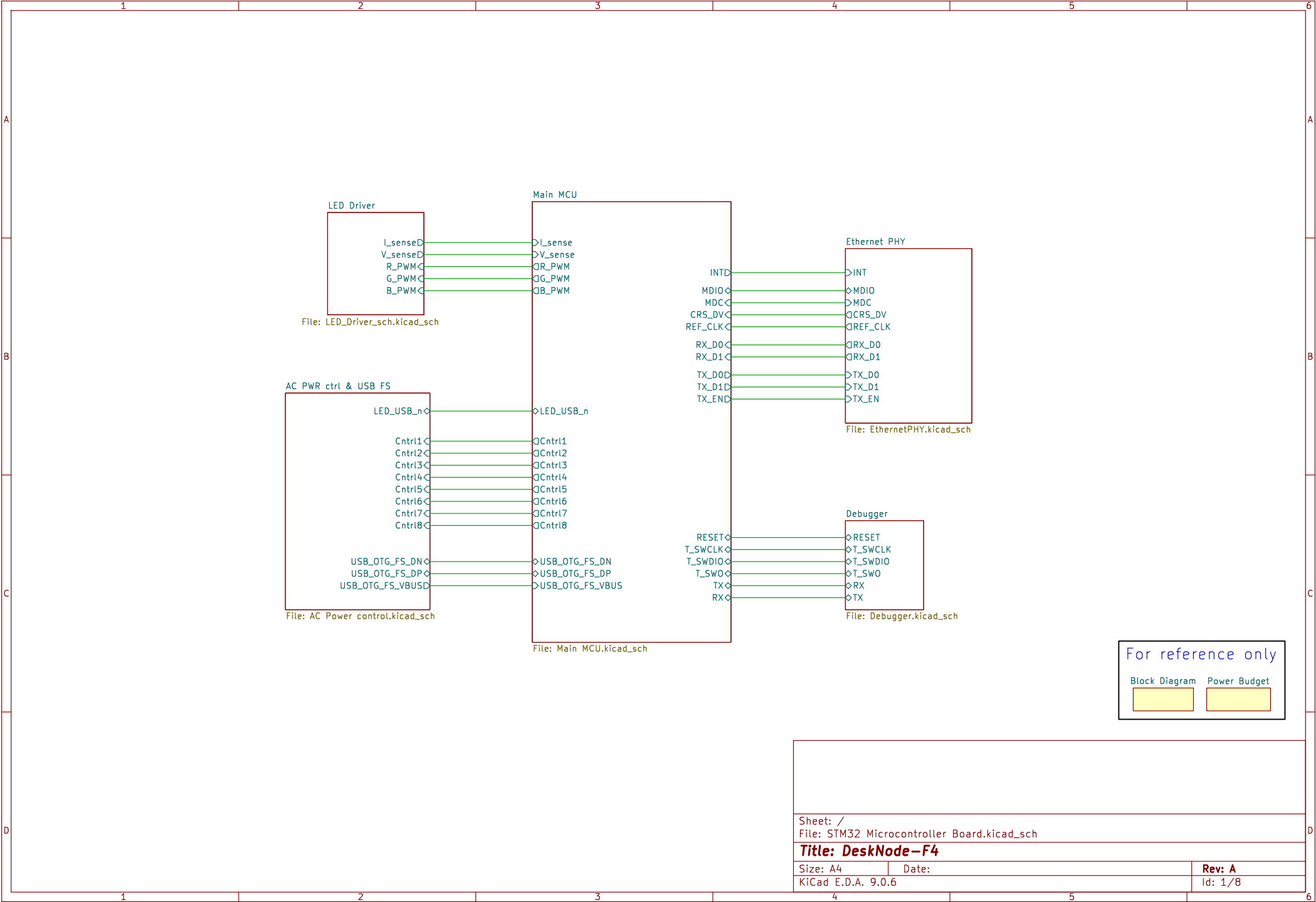




For reference only

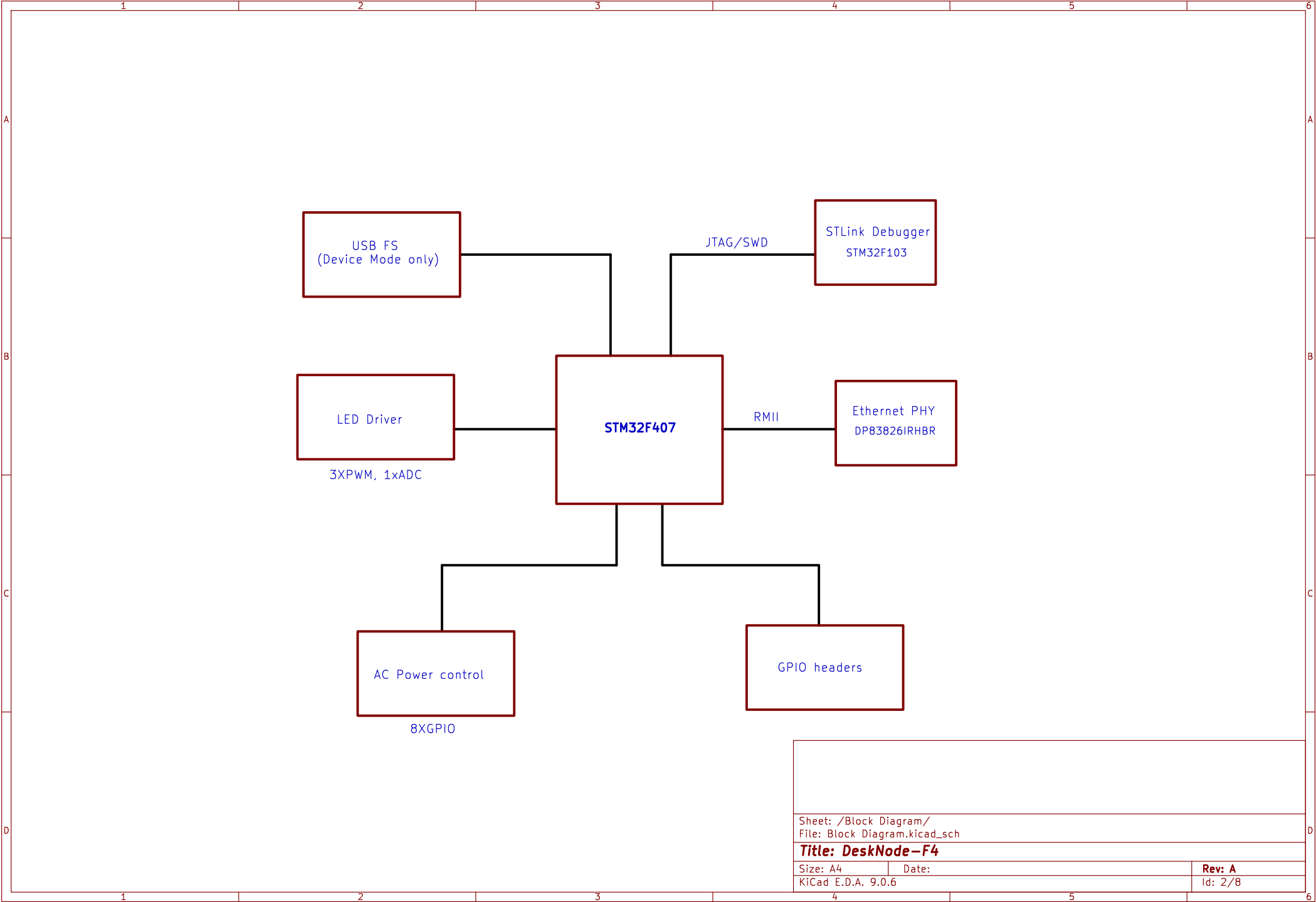
Block Diagram Power Budget

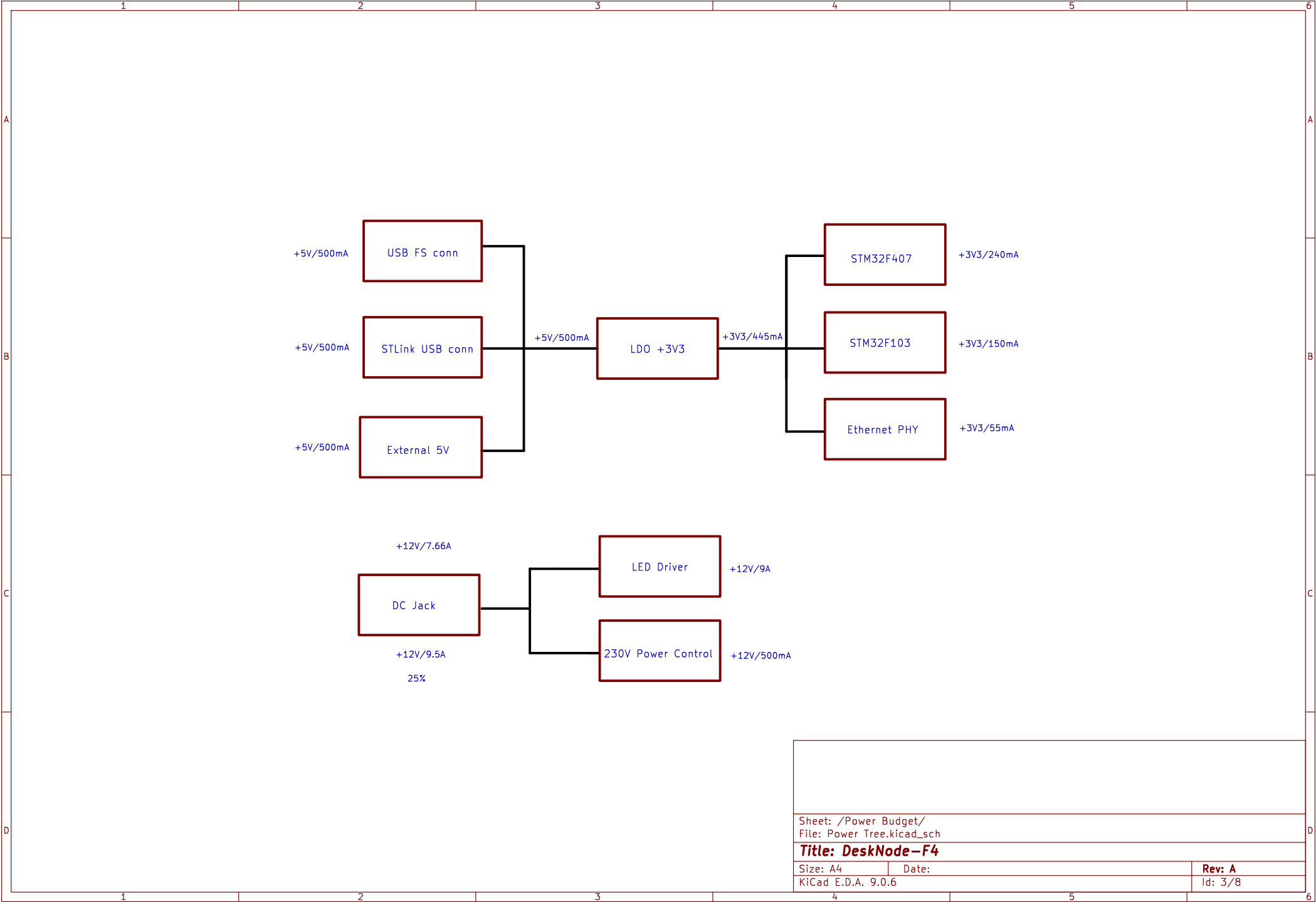


Sheet: /
File: STM32 Microcontroller Board.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	Rev: A
KiCad E.D.A. 9.0.6		Id: 1/8





Debugger/Programmer

Reset STM32F103

USB2.0 for Program/Debug

STM32F103 SWD

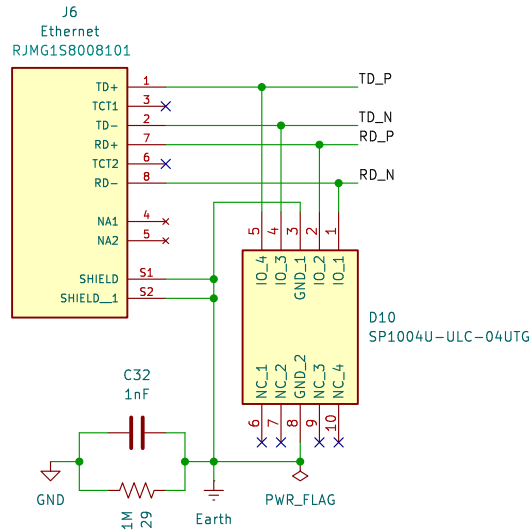
[illegible]

Diagram of the J4 header pin connections:

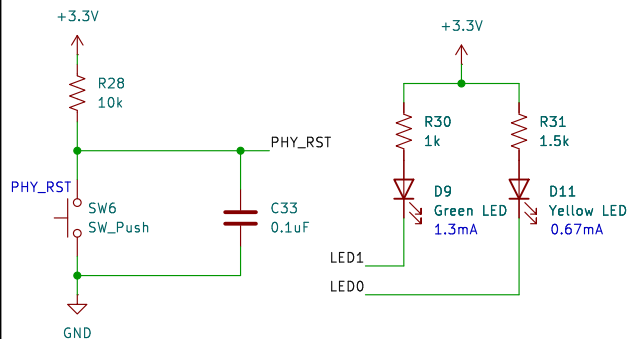
- Pin 1: +3.3V
- Pin 2: JTMS/SWDIO
- Pin 3: JTCK/SWCLK
- Pin 4: STM_RST
- Pin 5: TRACESWO
- Pin 6: GND

Title: DeskNode-F4		
Size: A4	Date:	Rev: A
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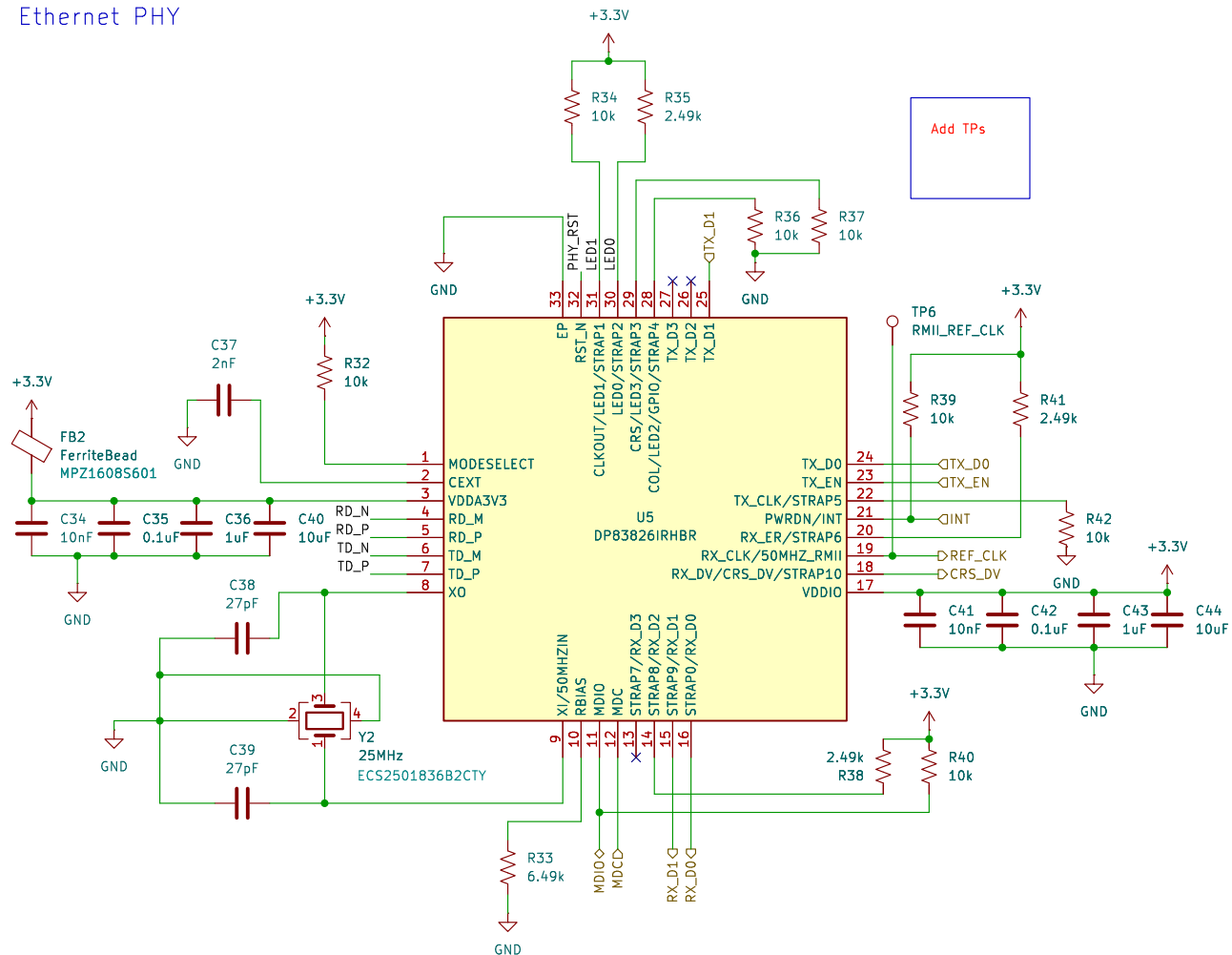
RJ45 Connector



ETH Reset and Status LEDs



Ethernet PHY



PHY Bootstrap table

Pin Name	Strap	Pin No	Strap Fn	Default	Resistor	Final State
RX_D0	Strap0	16	Auto-negotiation en/dis	0	-	0 - enabled
LED1	Strap1	31	Old Nibble en/dis	1	10k	1 - enabled
LED0	Strap2	30	PHY_ADD0	0	2.49k	1
LED3	Strap3	29	PHY_ADD1	0	10k	0
LED2	Strap4	28	PHY_ADD2	0	10k	0
TX_CLK	Strap5	22	RMII Leader/Follower	0	10k	0 - Leader
RX_ER	Strap6	20	Pin 31 is CLKOUT 25MHz/LED1	0	2.49k	1 - LED1
RX_D3	Strap7	13	Pin 18 is CRS_DV/RX_DV	0	-	0 - CRS_DV
RX_D2	Strap8	14	MII/RMII	0	2.49k	1 - RMII
RX_D1	Strap9	15	auto MDIX en/dis	0	-	0 - enabled
RX_DV	Strap10	18	MDIX/MDI (only applicable if auto-MDIX is dis)	0	-	0 - NA

Sheet: /Ethernet PHY/
File: EthernetPHY.kicad_sch

Title: DeskNode-F4

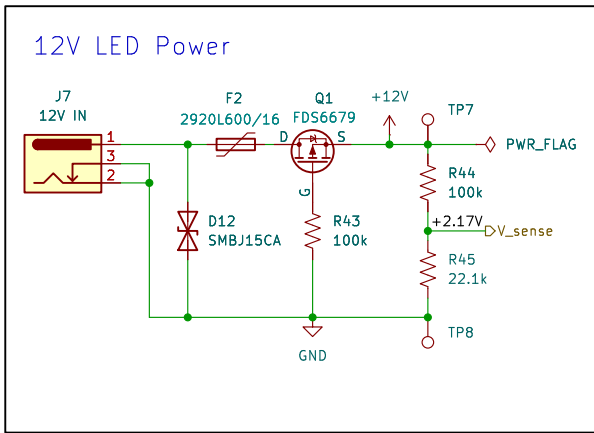
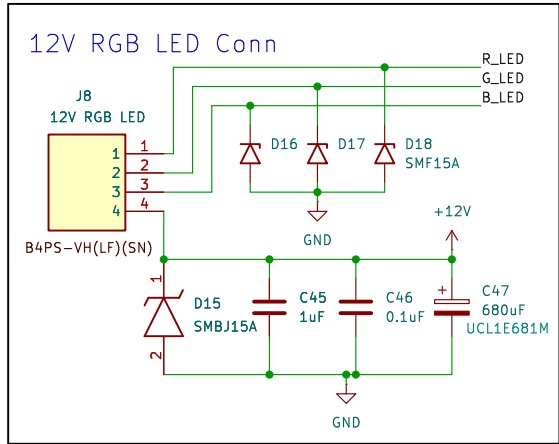
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Date:

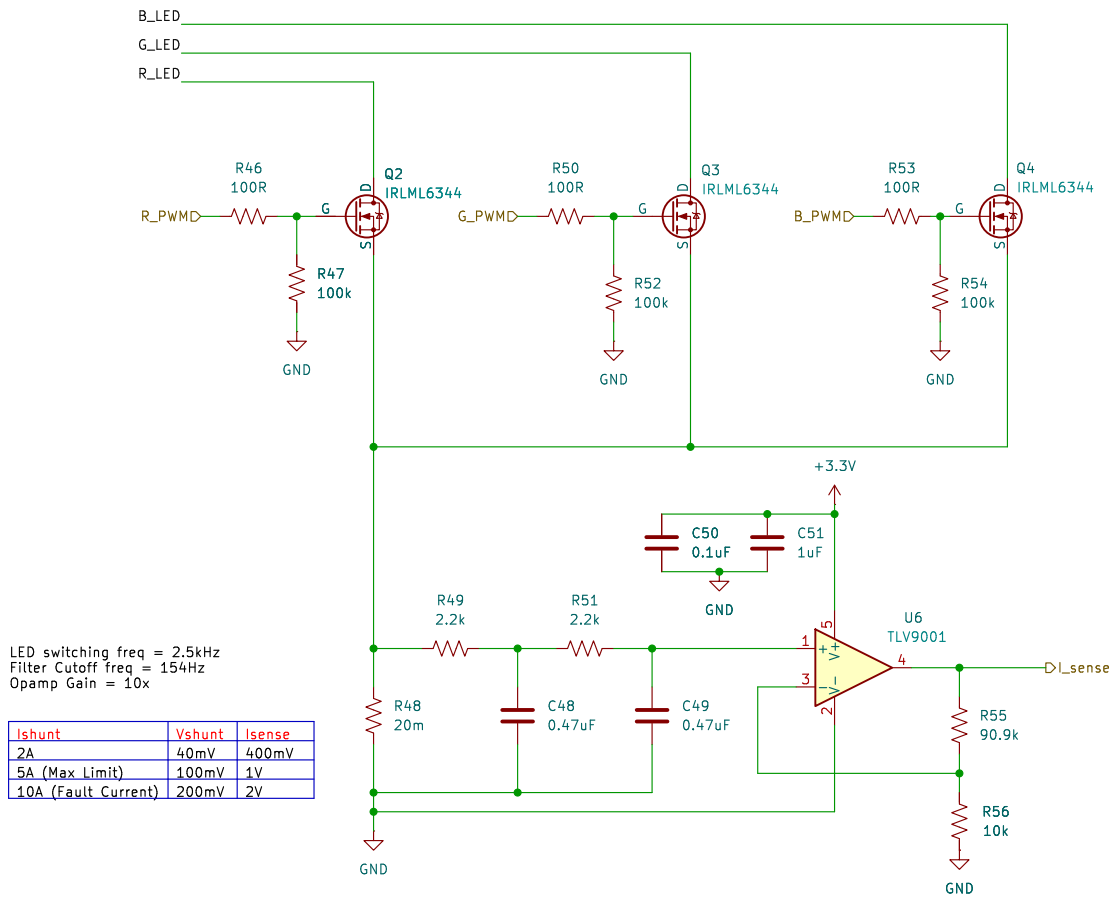
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Rev: A

Id: 6/8



LED Driver with current sensing



Sheet: /LED Driver/
File: LED_Driver_sch.kicad_sch

Title: DeskNode-F4

Size: A4

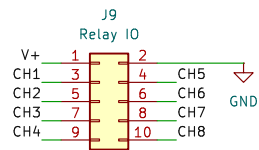
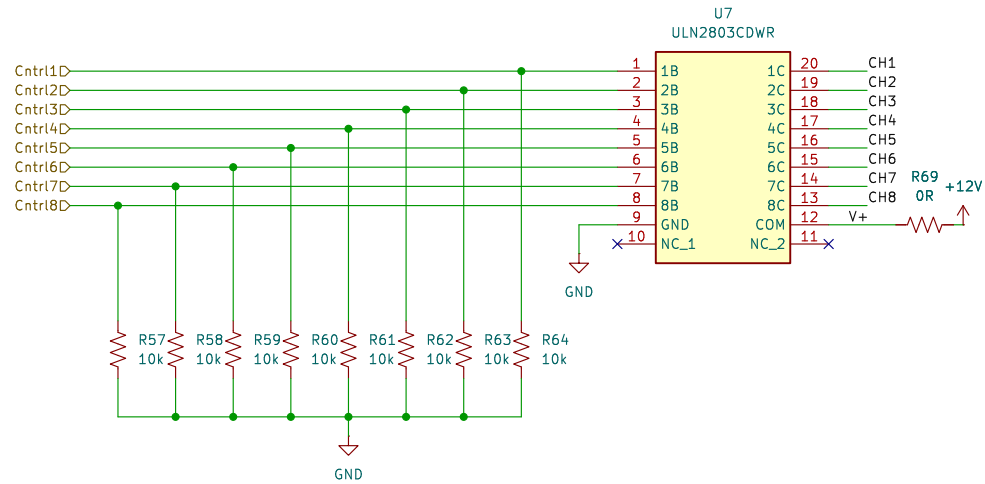
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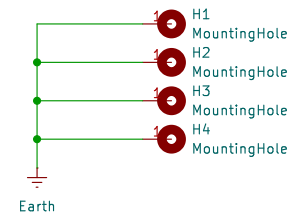
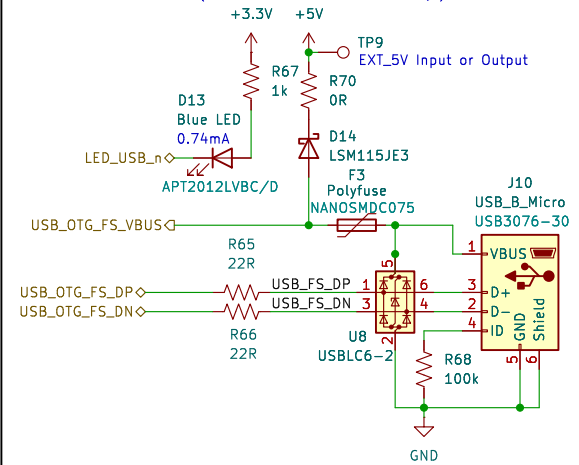
Rev: A

Id: 7/8

AC Power Control



USB FS OTG (Device mode only)



Sheet: /AC PWR ctrl & USB FS/
File: AC Power control.kicad_sch

Title: DeskNode-F4

Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev: A

Id: 8/8